AWS Academy Cloud Foundations Labs 1 to 3

By Albert Zhang

Background:

Lab 1:

In the first lab the user learns about **IAM (identity and access management):**
An IAM Define fine-grained access rights:
-Who can access the resource
-Which resources can be accessed and what can the user do to the resource
-How resources can be accessed
(IAM is a no-cost AWS account feature)

IAM Authorization:

Authorization is the process of determining what permission a user, service or application should be granted. After user has been authenticated, they must be authorized to access AWS services
By default, IAM users do not have permission to access any resource or data in an AWS account. Instead, explicit permission needs to be granted to a user, group or role by creating a policy, which is a document in JSON format. A policy list permission that allows or denies access to resources in the AWS account.
This lab demonstrates how to access pre-created IAM Users and groups by inspecting IAM policies as applied to the pre-created groups, this lab also follows a real-world scenario, teaching the

Lab 2:

Amazon Virtual Private Cloud (Amazon VPC):
The Amazon VPC lets someone launch Amazon Web Services resources in a virtual network that is customizable. The virtual network is very similar to traditional network, with the difference being that Amazon VPC can be scalable using the infrastructure of AWS. VPCs can be created to cover multiple Availability Zones

The second lab shows the user how to use Amazon Virtual Private Cloud (VPC) to create their own VPC and add more components to produce a customized network. This lab also demonstrates how to create a security group and customize an EC2 instance which can run a web server, with the EC2 instance running a subnet in the VPC.

After someone completes this lab, they should be able to create a VPC, create subnets, configure a security group, and finally be able to launch an EC@ instance into a VPC.

Lab 3:

Amazon Elastic Computer Cloud (EC2):

EC2 is a web service that provides resizable computing capacity in cloud. EC2 is designed to make web-scale cloud computing easier for developers.

Amazon EC2 has a simple web service interface that lets you obtain and configure capacity easily. It provides you with complete control of your computing resources and lets you run on Amazon's proven computing environment. Amazon EC2 reduces the time needed to get and boot new server instances to minutes, enabling you to quickly scale capacity either up or down, as your needs change.

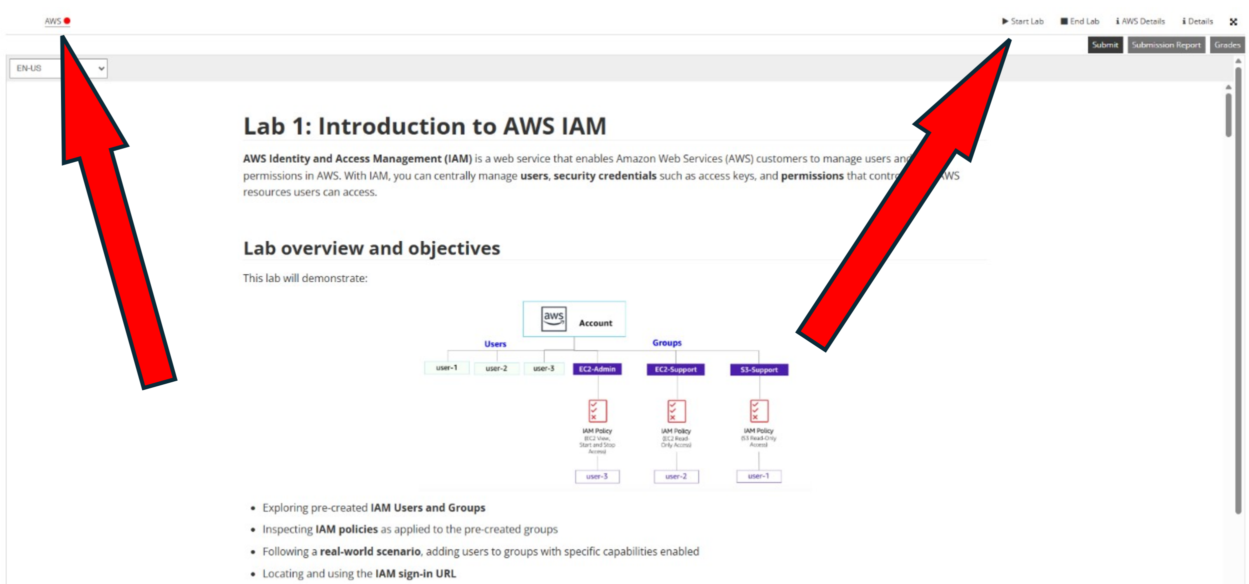
Amazon EC2 completely changes the economics of computing by allowing the user to pay only for capacity that one needs. Amazon EC2 gives developers the tools to build failure resilient applications and separate themselves from common failure scenarios

This lab provides the user with a basic overview of launching, resizing, managing and monitoring an Amazon EC2 instance. After completing the lab, one should be able to launch a web server with termination protection enabled, monitor your EC2 instance, modify security group that one's web server is using, resize Amazon EC2 instance to scale, enabling stop protection, and finally stopping the EC2 instance.

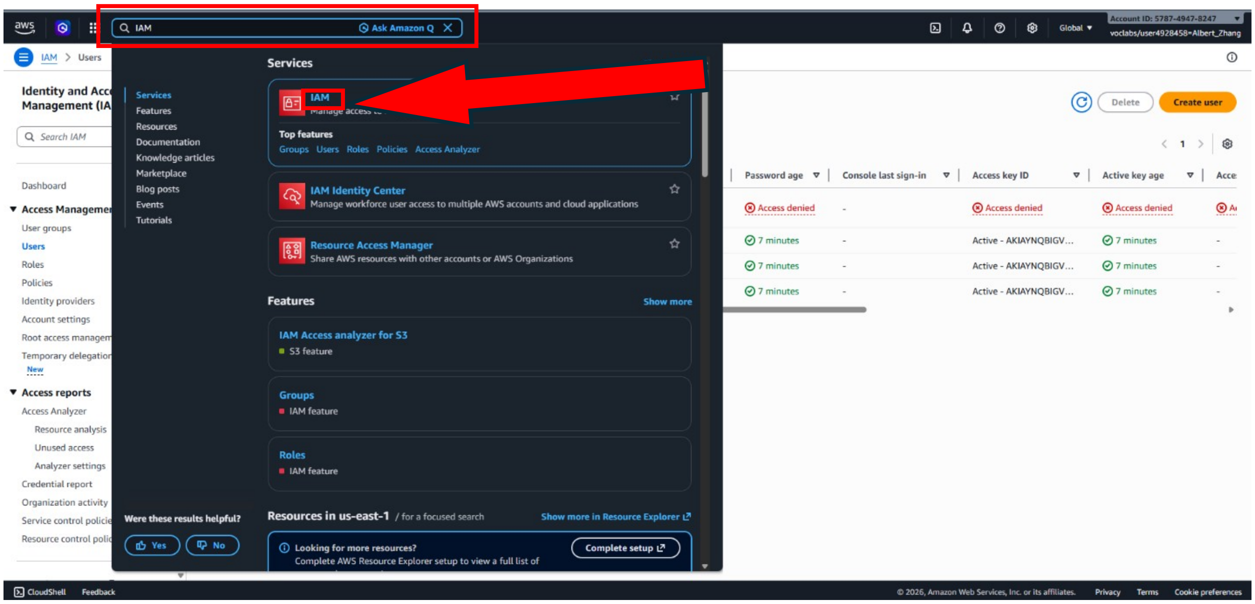
Configurations:

Lab 1:

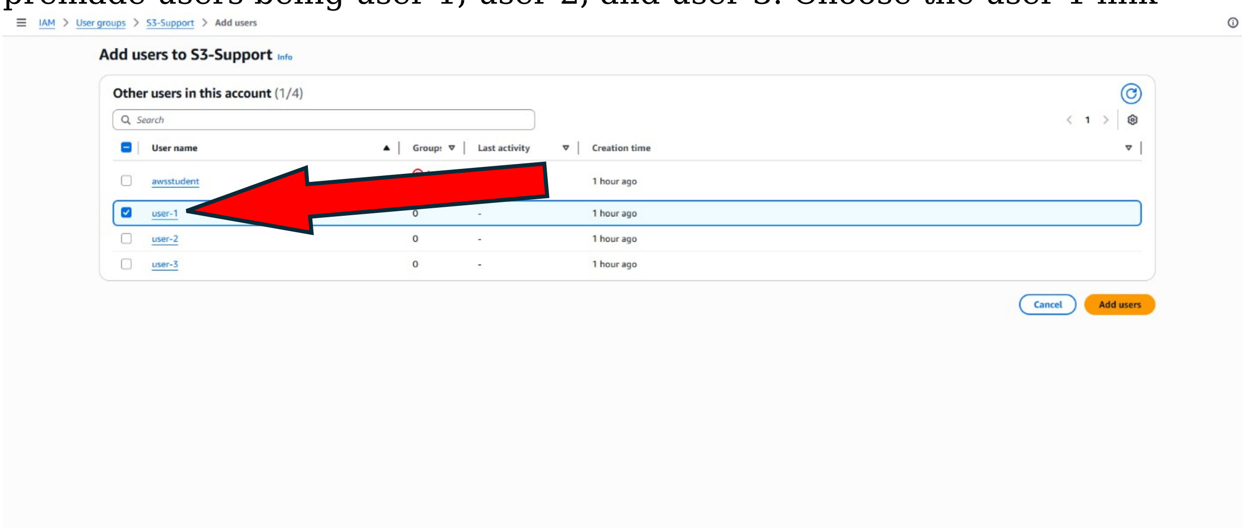
Star with clicking "Start Lab" and then the AWS link



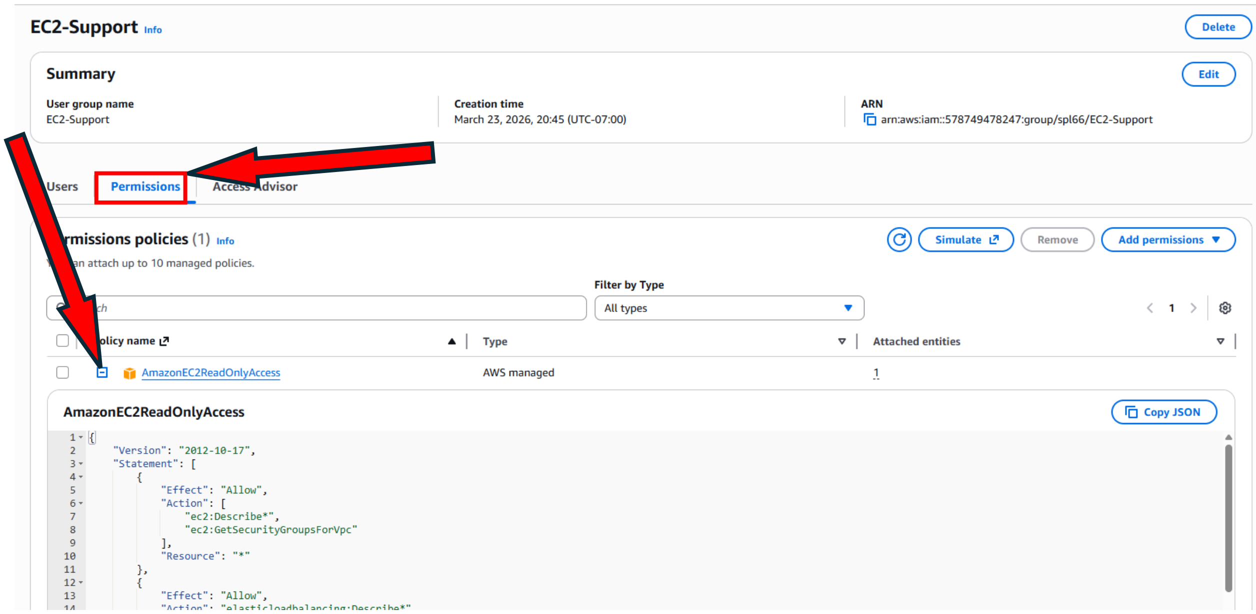
Now you will be brought to the AWS page, go to the right of Services and search IAM to open the IAM console



Go to the navigate pane on the left and choose “Users”, you will see the 3 premade users being user-1, user-2, and user-3. Choose the user-1 link

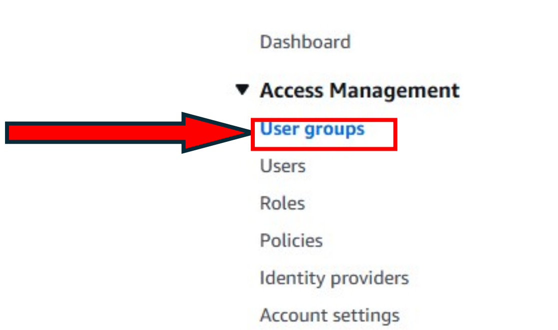


After going through the user-1 link click “Permissions”

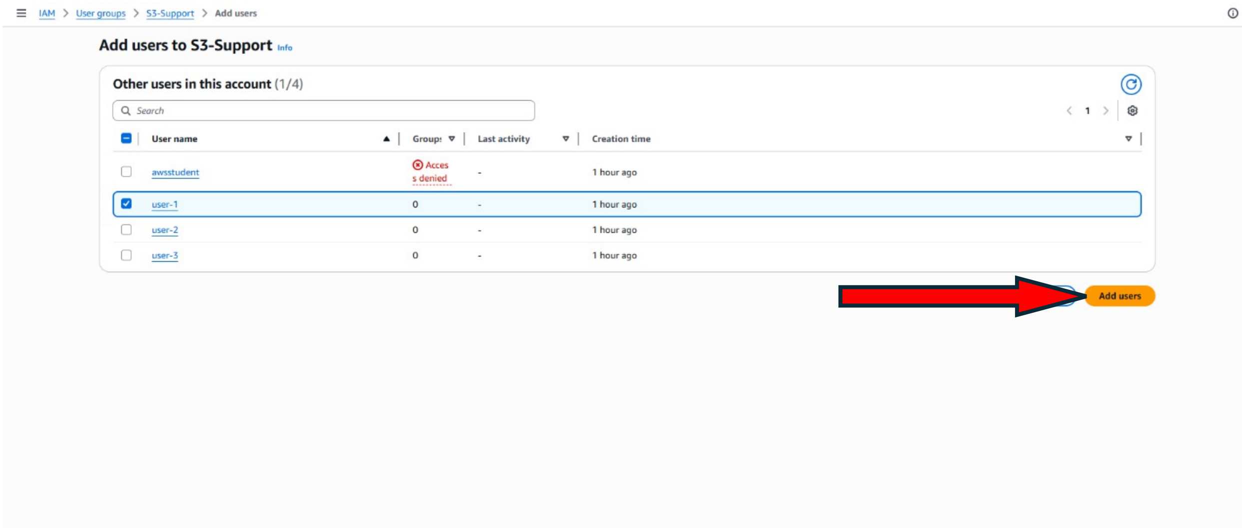


You will see that this group is running something called "AmazonEC2ReadOnlyAccess", click the plus(+) icon next to it to view the policy details. Repeat the same steps but for user-2 and 3.

Now you will add users to groups, go to the navigation pane again and choose "User groups this time.

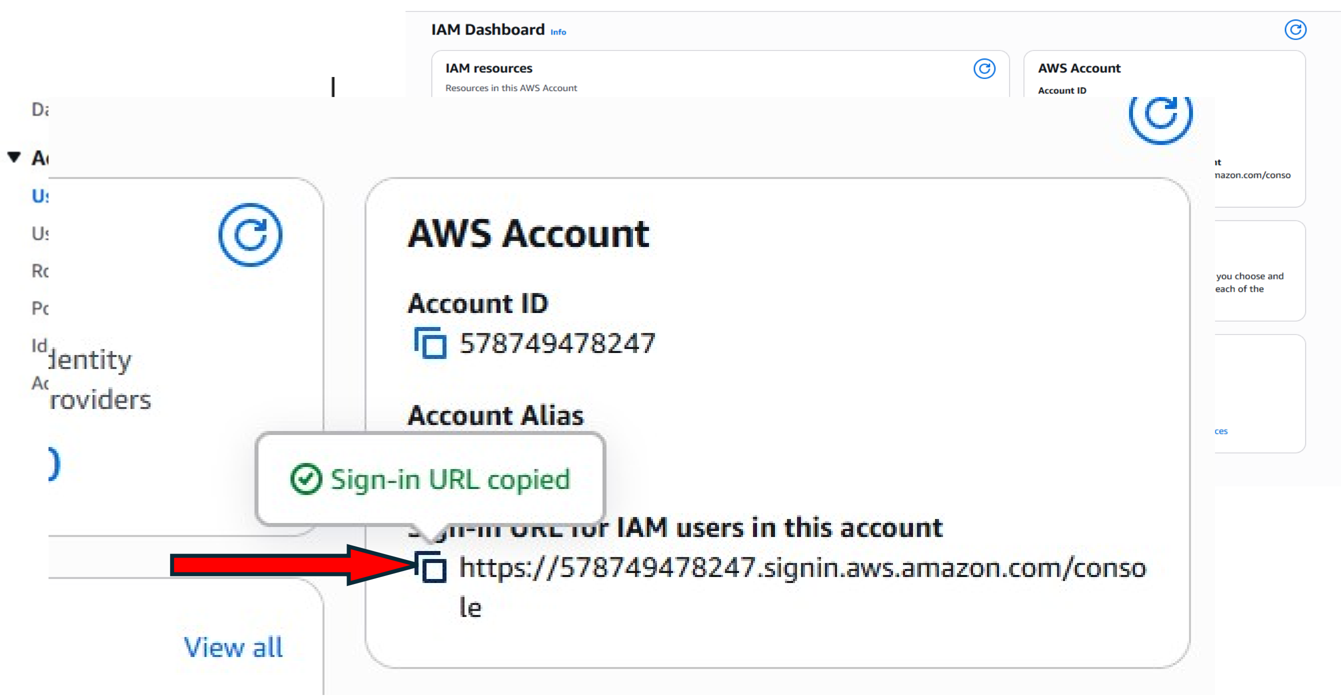


Choose the S3-Support group link and choose the Users tab, after that choose "Add users", add user-1 to the group



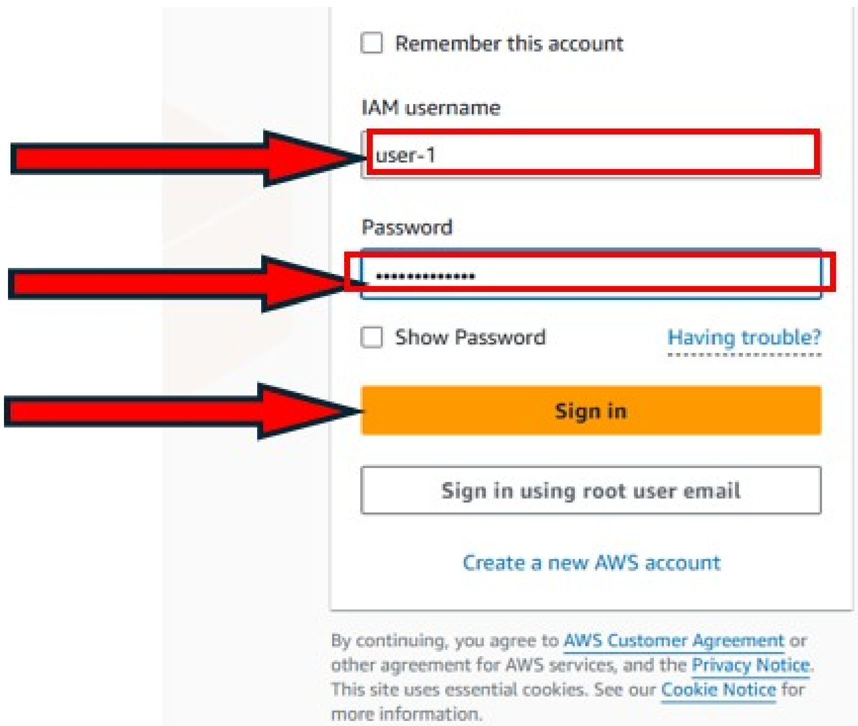
Repeat the same steps for user-2 and user-3 but adding User-2 to the “EC2-Support Group” and user-3 to the “EC2-Admin Group”. When done successfully you will see this notification:

Next, you will sign-in and test users. On the navigation pane to the left, choose “Dashboard”, you will see a sign-in URL for IAM users in this account link is displayed on the right. The link can be used to sign-in to AWS account you are currently using.



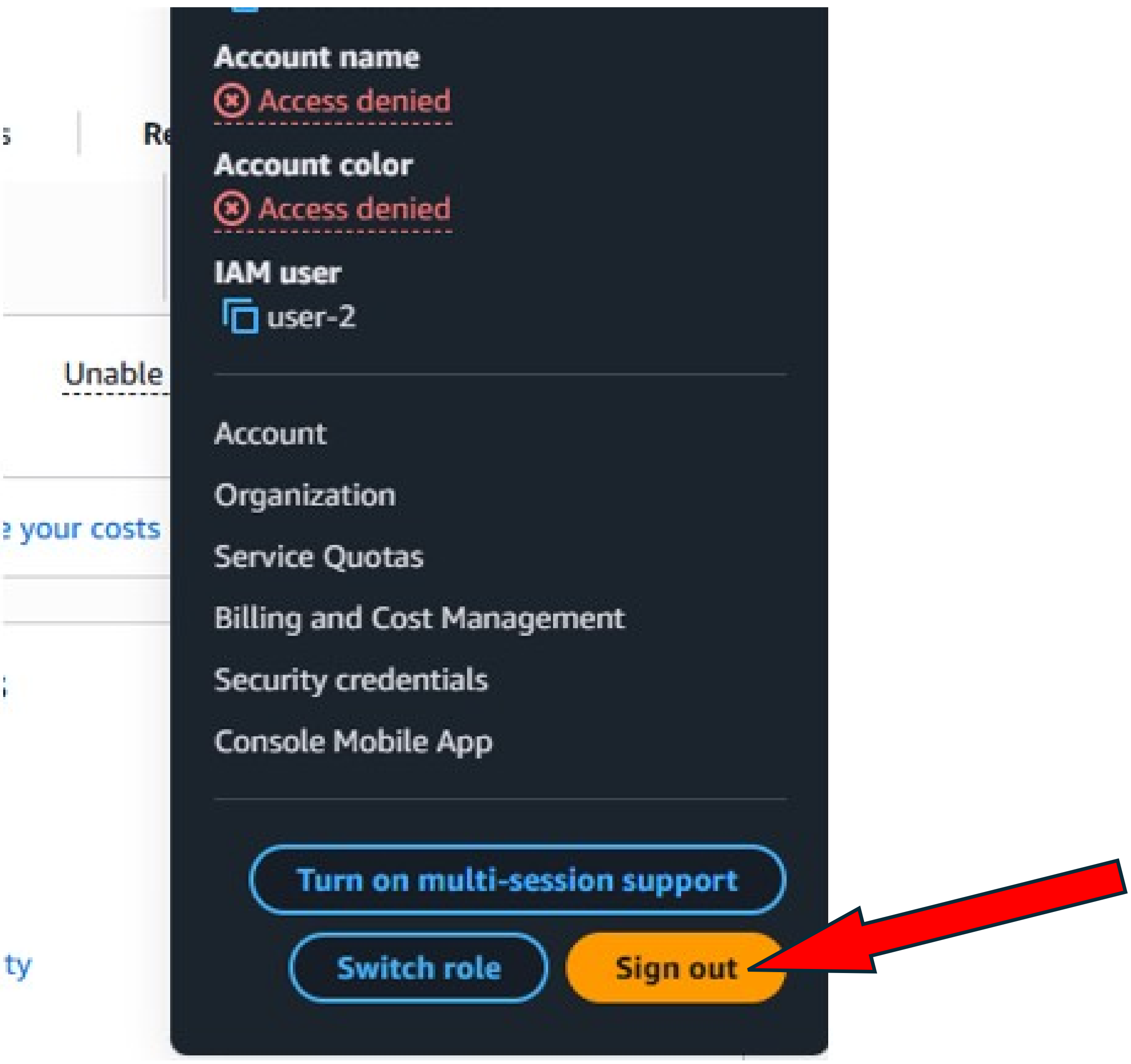
Next, copy the sign-in URL for IAM users in this account

Open a new private window and paste the IAM user sign-in link you just copied. You will be brought to this page with your account ID or alias already entered in.
You will need to enter in the IAM username and password.
User-1:
Username: user-1, password: Lab-Password1
User-2:
Username: user-2, password: Lab-Password2
User-3:
Username: user-3, password: Lab-Password3

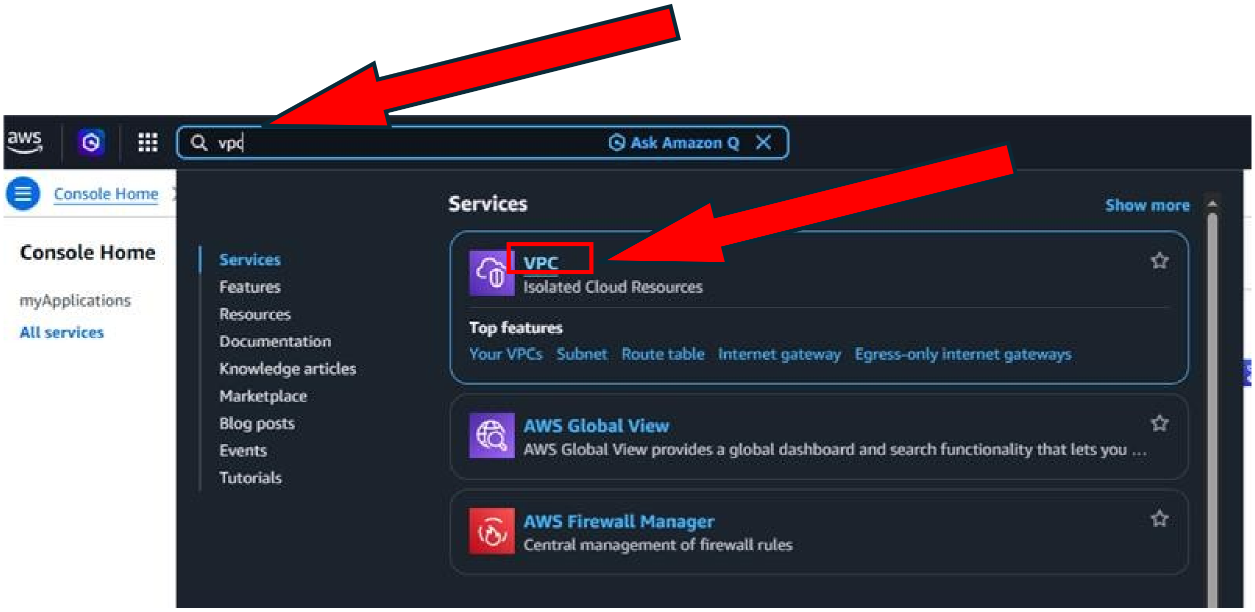


After signing in to the IAM user accounts, choose instances, and in the instance state menu select “Stop instance”

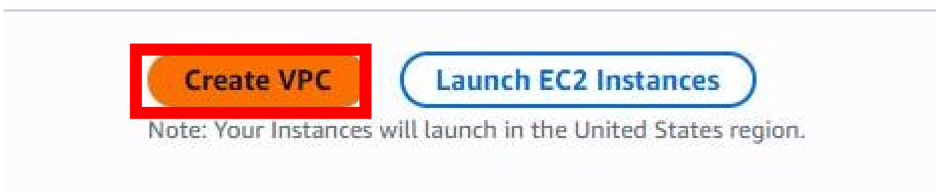
User-1: search for S3 in services, this account has permission to view a list of Amazon S3 buckets and the contents, after completing the steps go to the top of the screen, chooser user-1, 2 or 3 and click “Sign out”



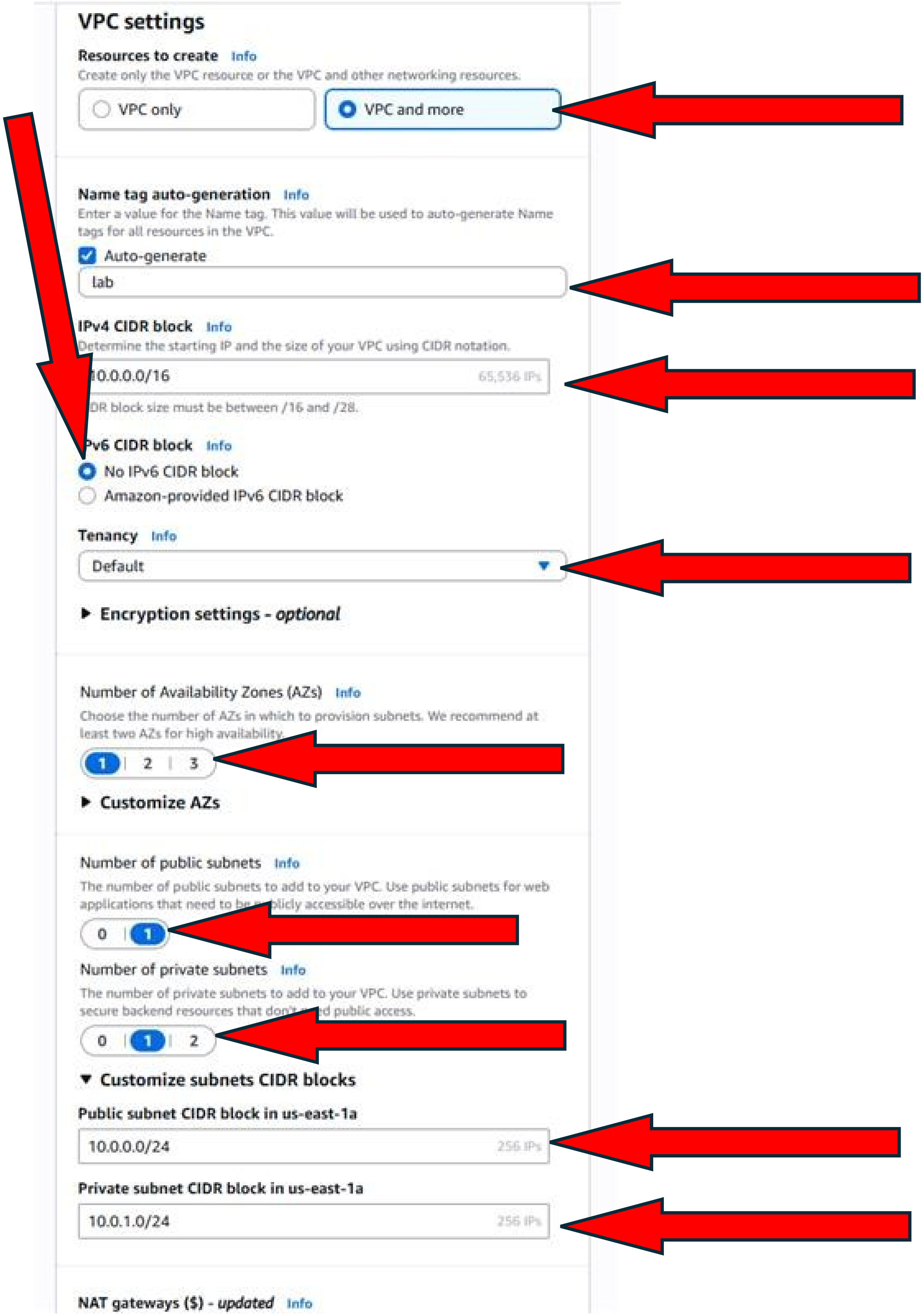
Lab 2:
In the search box to the right of Services search for “VPC” and open the VPC console



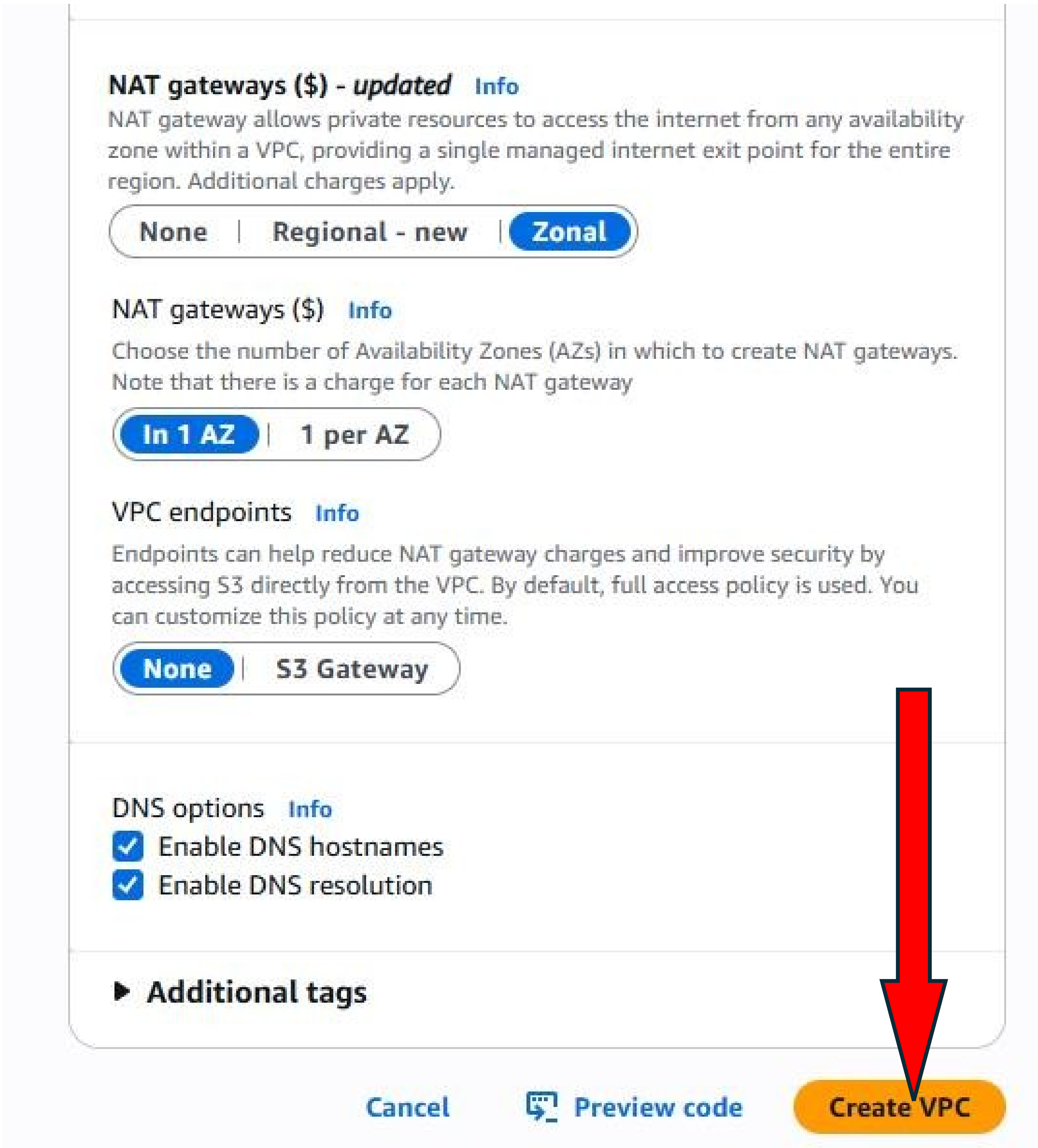
Next, begin creating a VPC. Select "Create VPC"



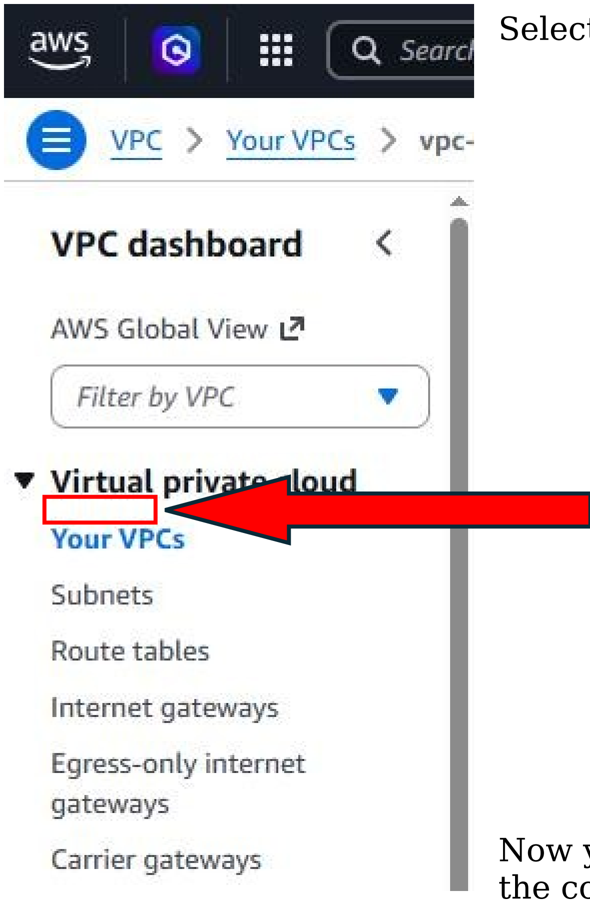
Choose VPC and more.
Under Name tag auto-generation, keep Auto-generate selected, however change the value from project to lab.
Keep the IPv4 CIDR block set to 10.0.0.0/16
For Number of Availability Zones, choose 1.
For Number of public subnets, keep the 1 setting.
For Number of private subnets, keep the 1 setting.
Expand the Customize subnets CIDR blocks section.
Change Public subnet CIDR block in us-east-1a to 10.0.0.0/24Change Private subnet CIDR block in us-east-1a to 10.0.1.0/24Set NAT gateways to In 1 AZ.
Set VPC endpoints to None.
Keep both DNS hostnames and DNS resolution enabled.
Your configurations should look like:



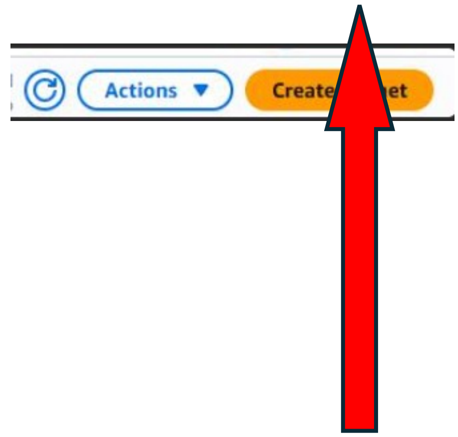
After that select “Create VPC” at the bottom of the screen:



Next you will create a subnet, on the navigation pane, select “Subnets”

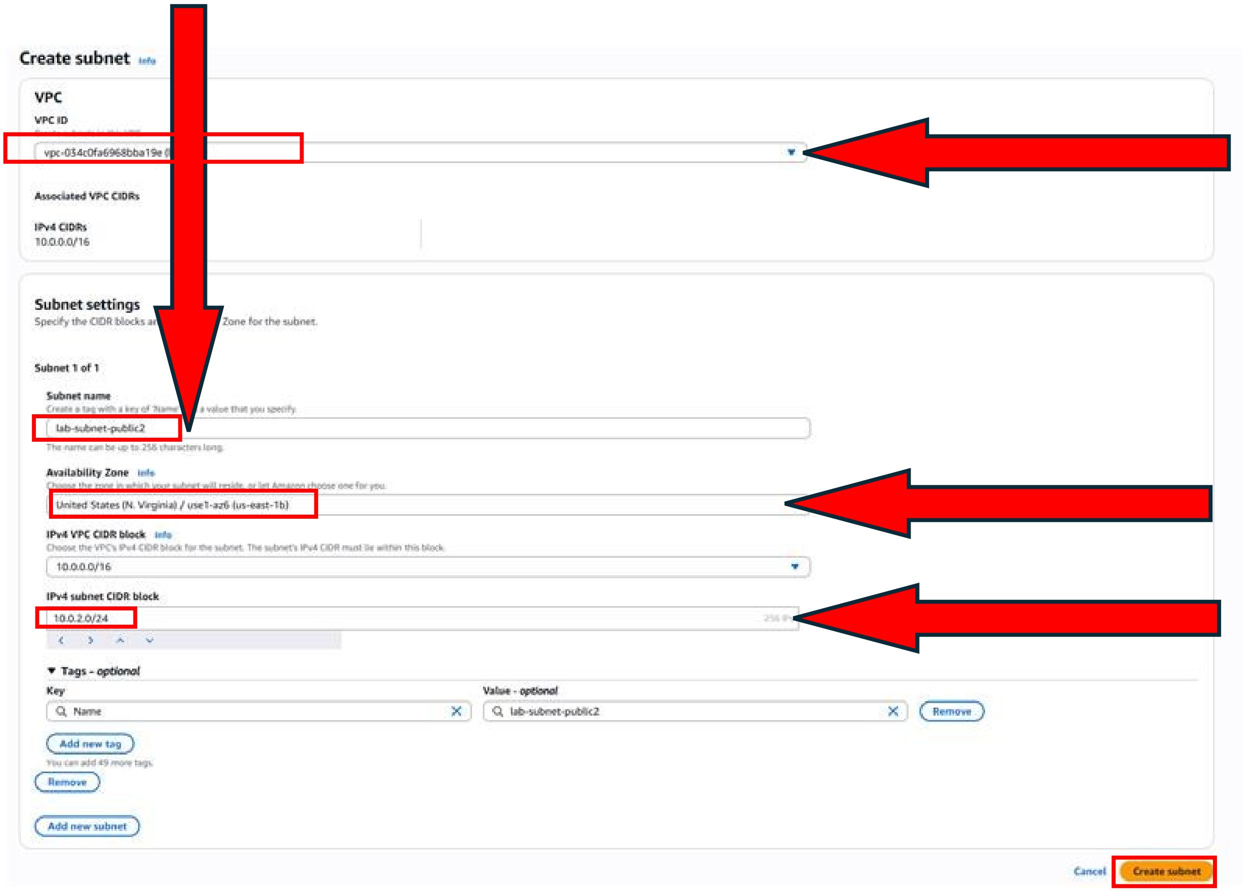


Select “Create Subnet”, next to Actions



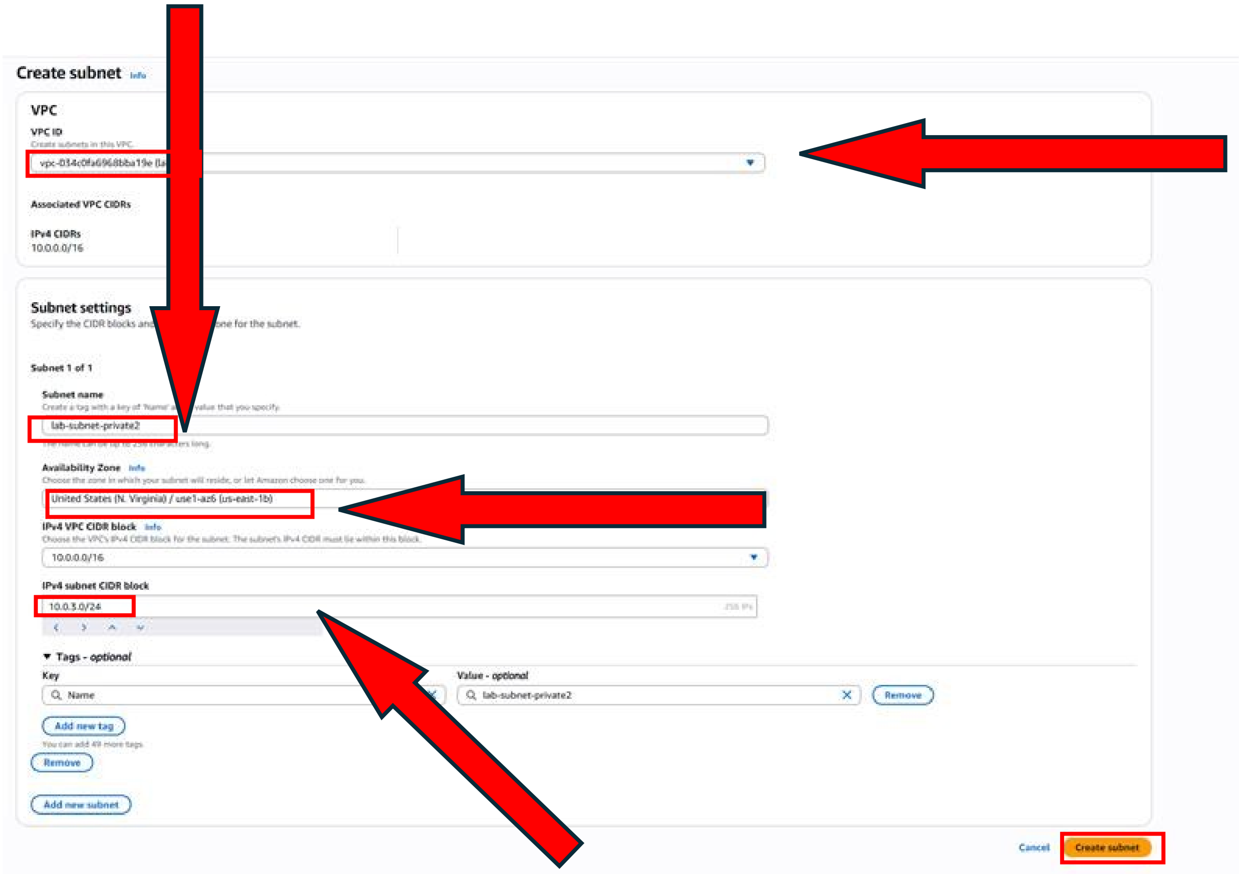
Now you will configure 2 new subnets with the configurations:

VPC ID: lab-vpc (select from the menu).
Subnet name: lab-subnet-public2
Availability Zone: Select the second Availability Zone (for example, us-east-1b)
IPv4 CIDR block: 10.0.2.0/24

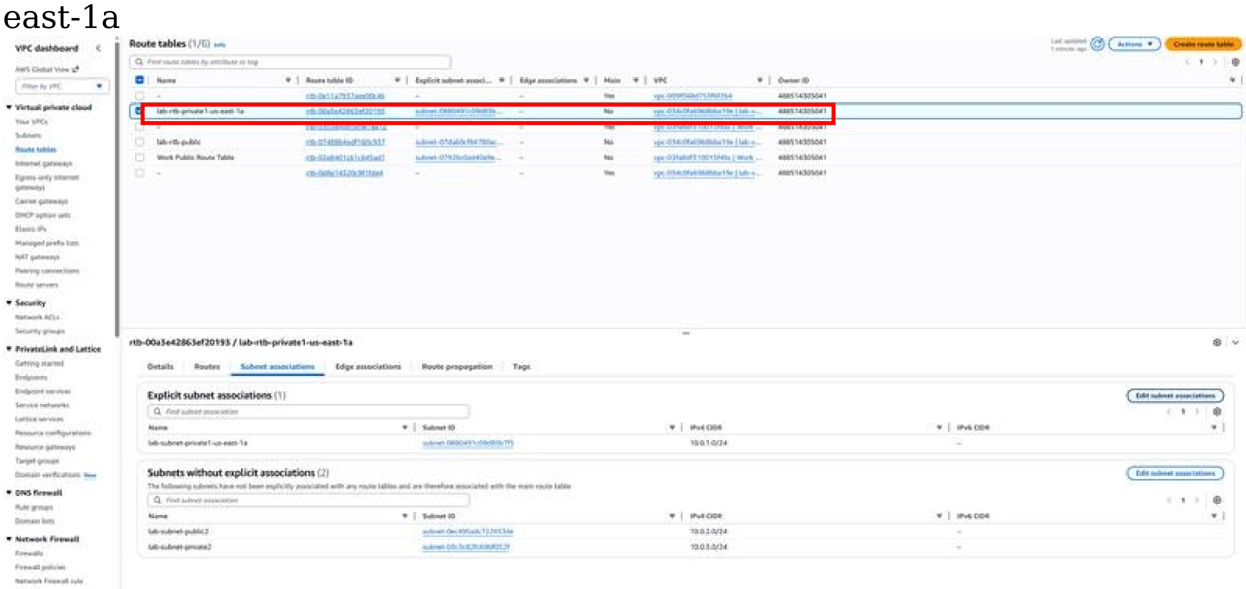


Followed by the next subnet:

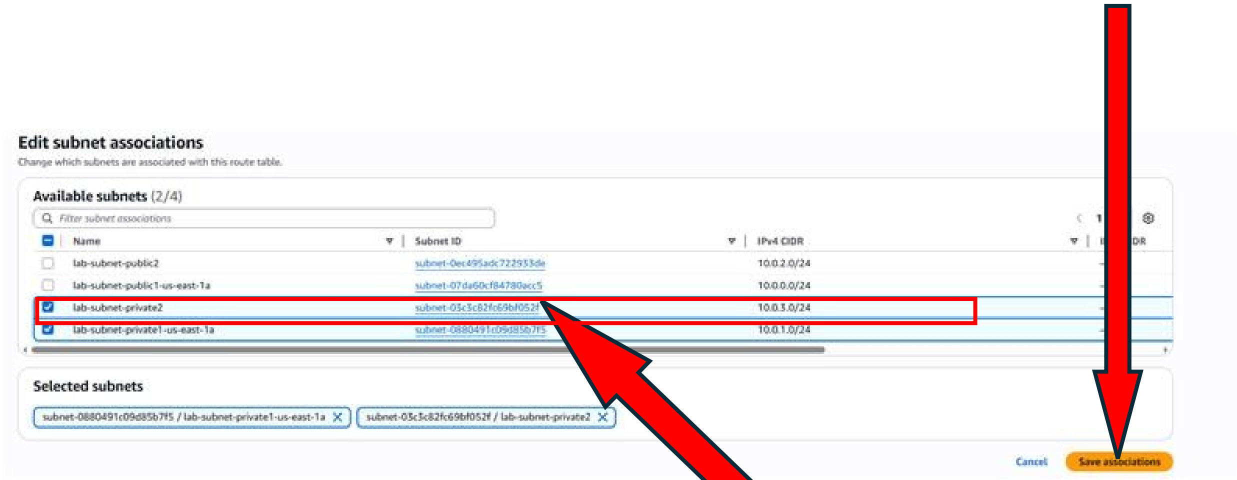
VPC ID: lab-vpc
Subnet name: lab-subnet-private2
Availability Zone: Select the *second* Availability Zone (for example, us-east-1b)
IPv4 CIDR block: 10.0.3.0/24



At the left navigation pane choose “Route Tables” , select lab-rtb-rivate1-us-east-1a



In explicit subnet associations panel, choose “Edit subnet associations”



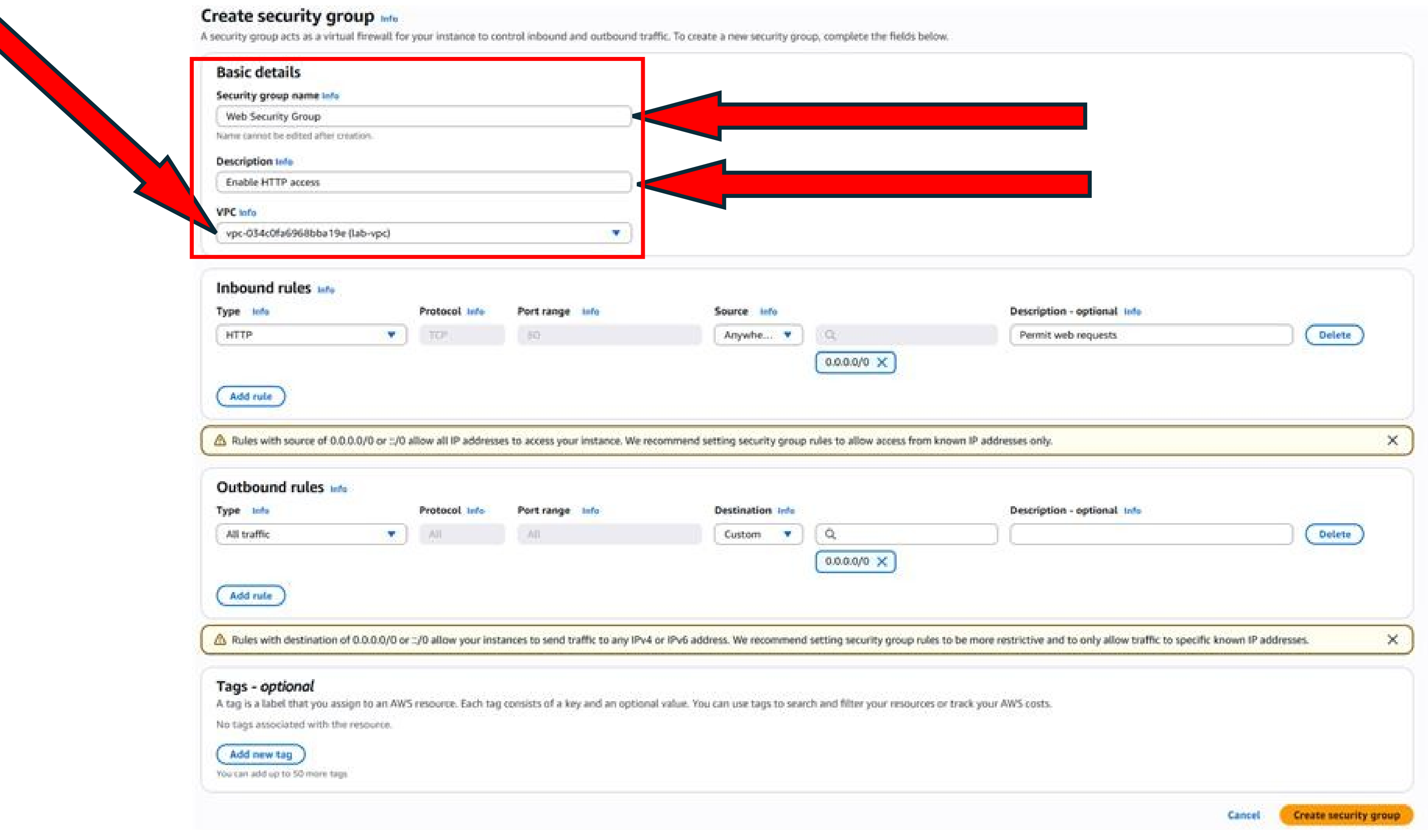
Keep lab-subnet-private-us-east-1a selected but also select “lab-subnet-private2”. Choose “Save associations”

Now you will create a VPC security group, acting like a firewall.

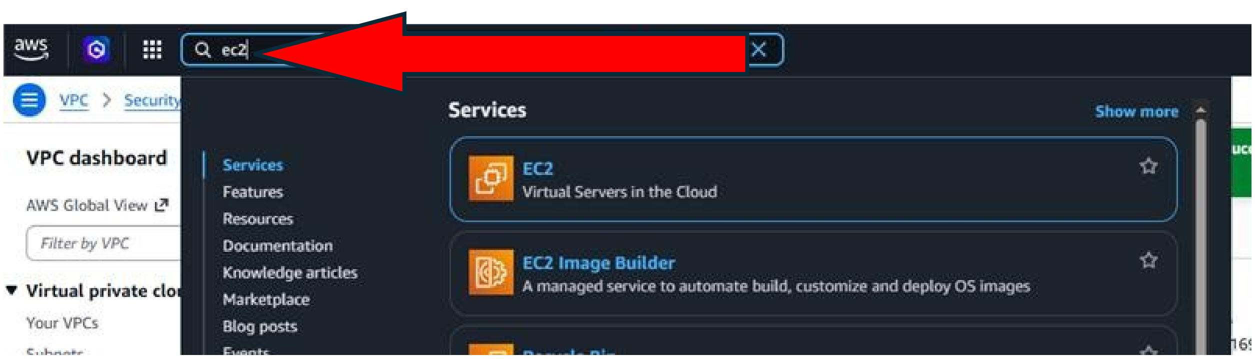
On the left navigation pane choose “Security groups”, then choose “Create security groups” and then configure:

Security group name: web Security Group
Description: Enable HTTP access
VPC: choose the x to remove the currently selected VPC, then from the drop-down list choose “lab-vpc”

It should look something like:

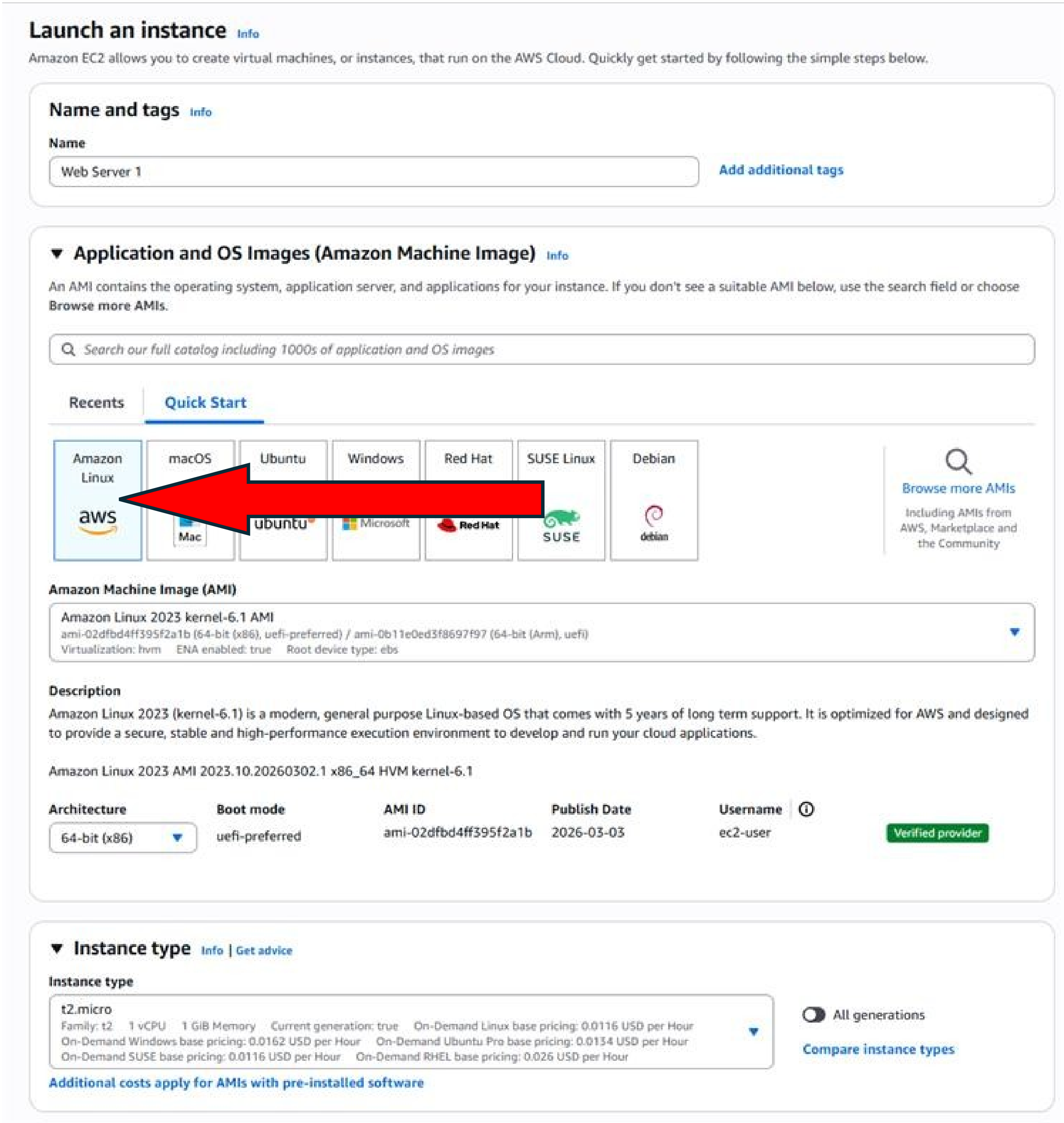


Now you will launch an Amazon EC2 instance into the new VPC. In Services search for and choose EC2 and open the EC2 console



From the launch instance menu choose “Lauch instance”, give the instance give it the name Web Server 1.

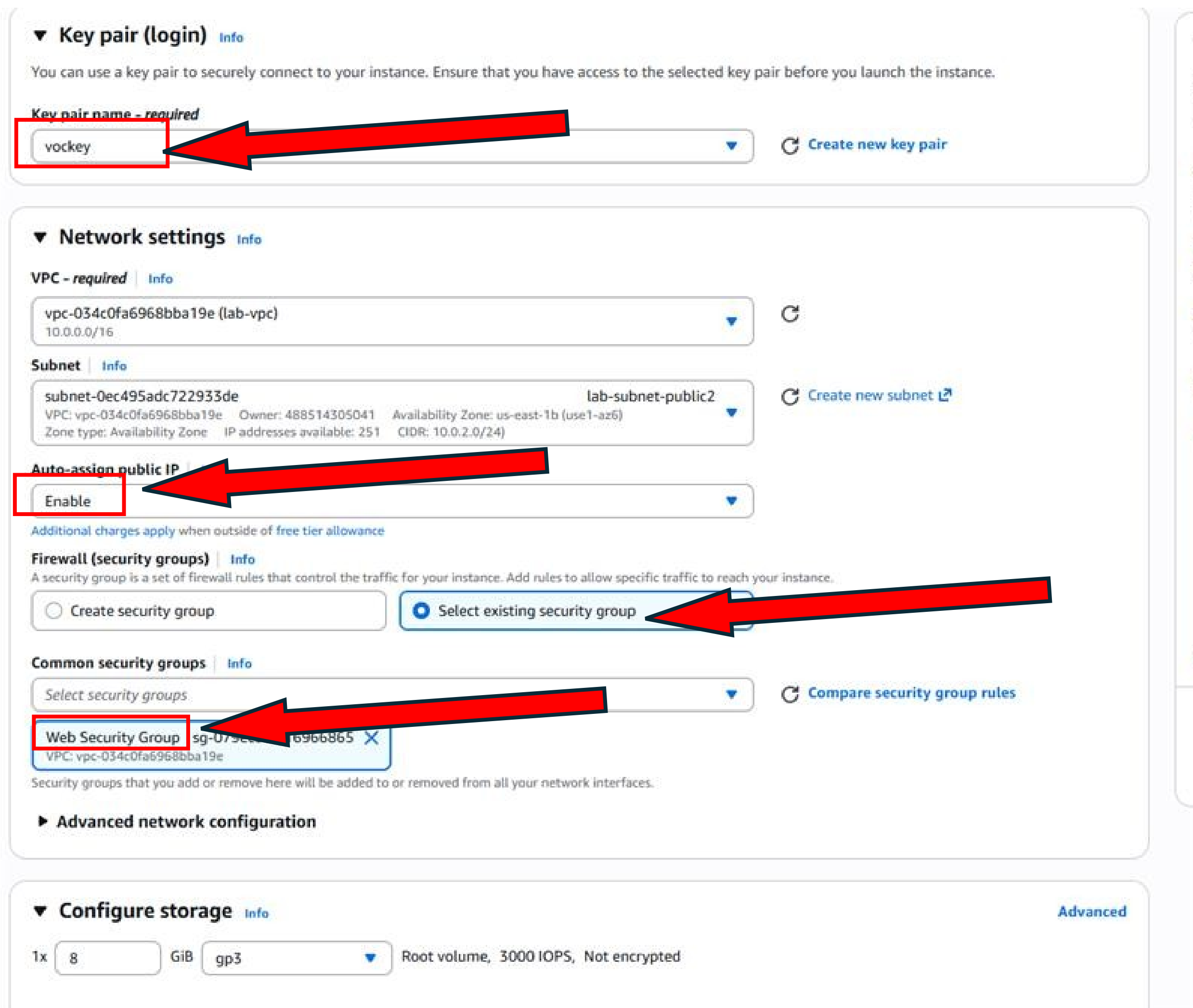
Select the interface type t2.micro.



From the key pair name menu select "vockey" and choose "Edit" then configure":
Network: lab-vpc
Subnet: lab-subnet-public2
Auto-assign public IP: Enable

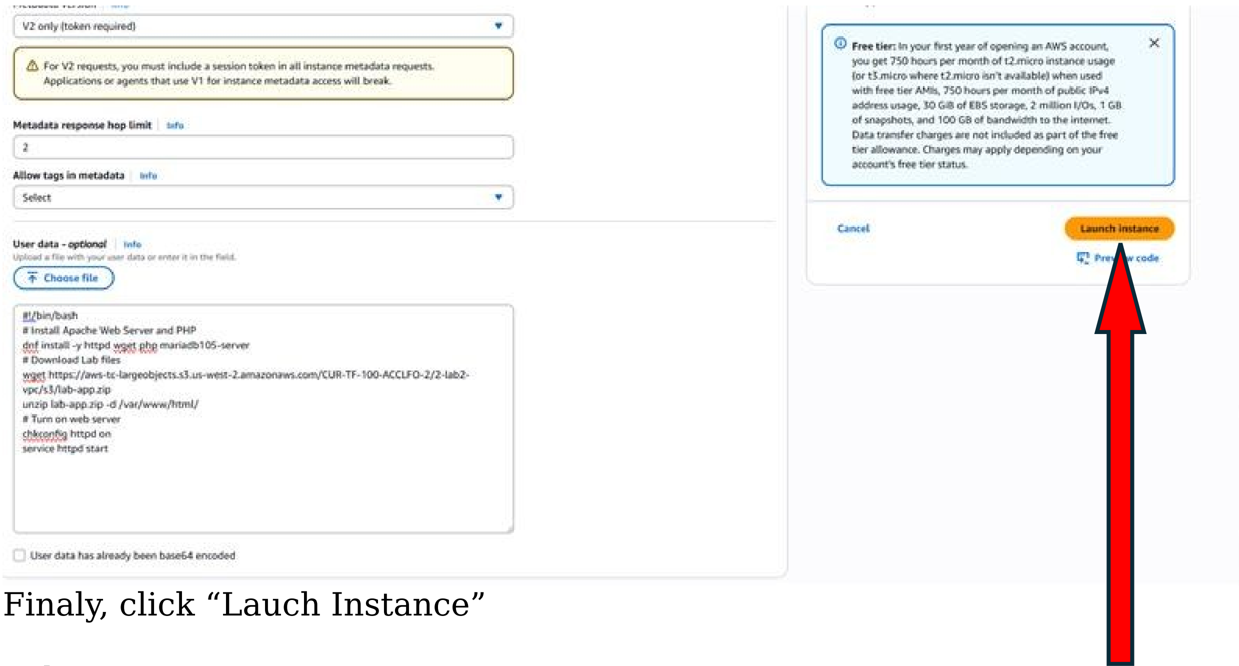
Under firewall, choose "Select existing security group"
Under common security groups, choose "Web Security Group"

It should look like:



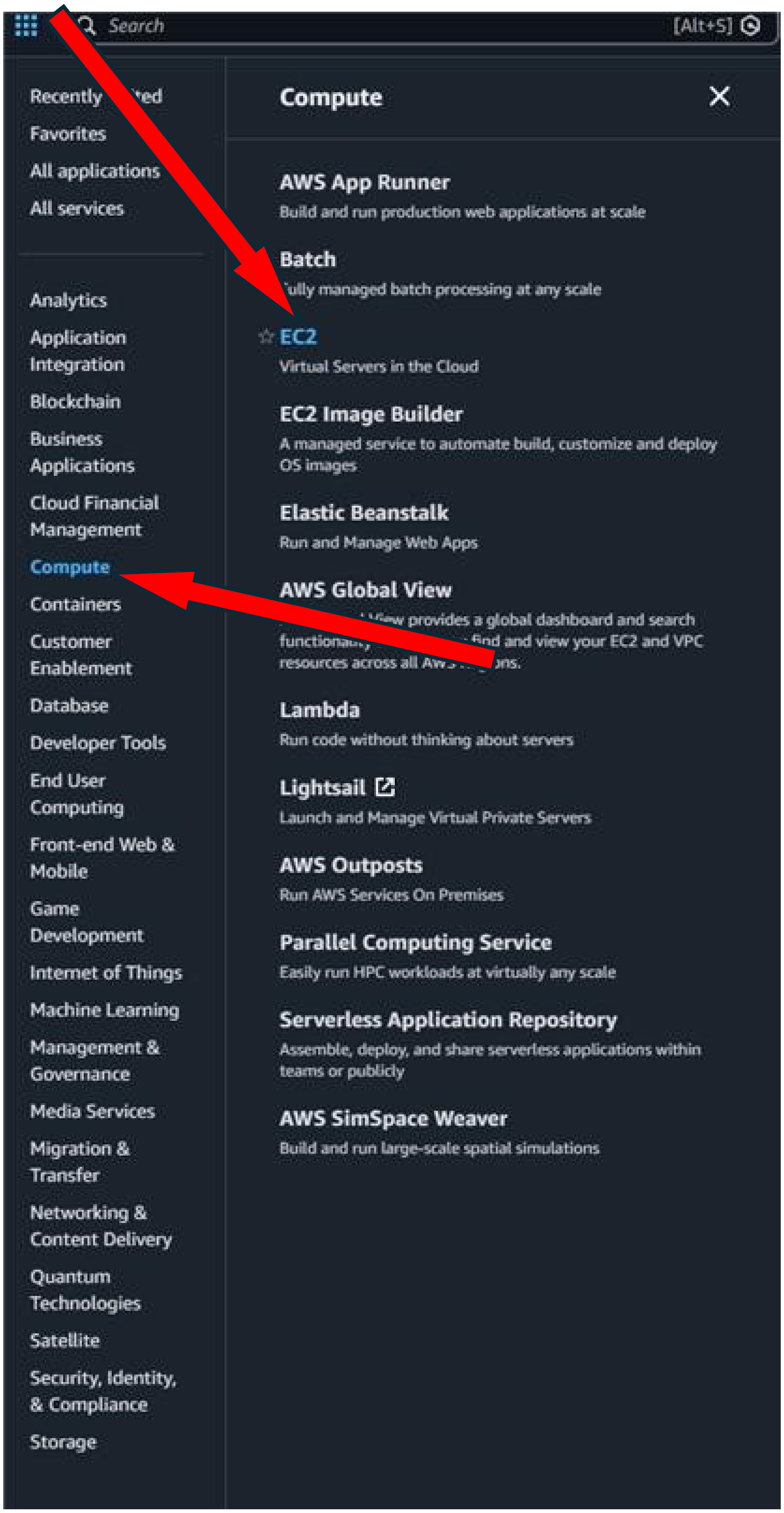
Expand the “Advanced details” panel
And paste this code into the user data box:

```
!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

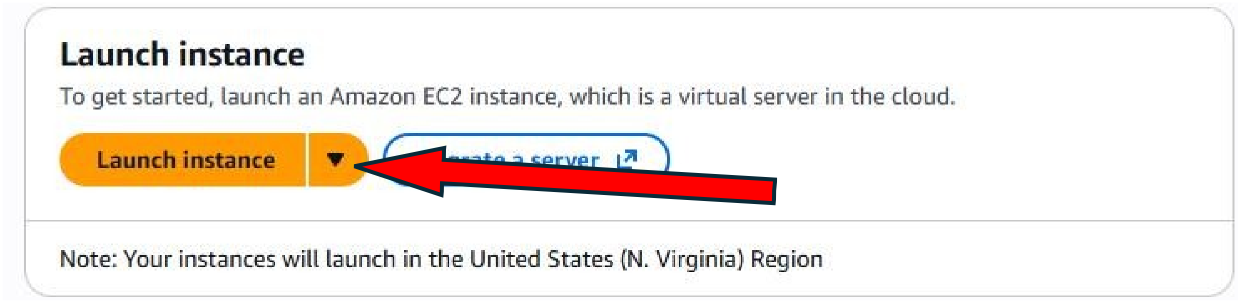


Finaly, click “Lauch Instance”

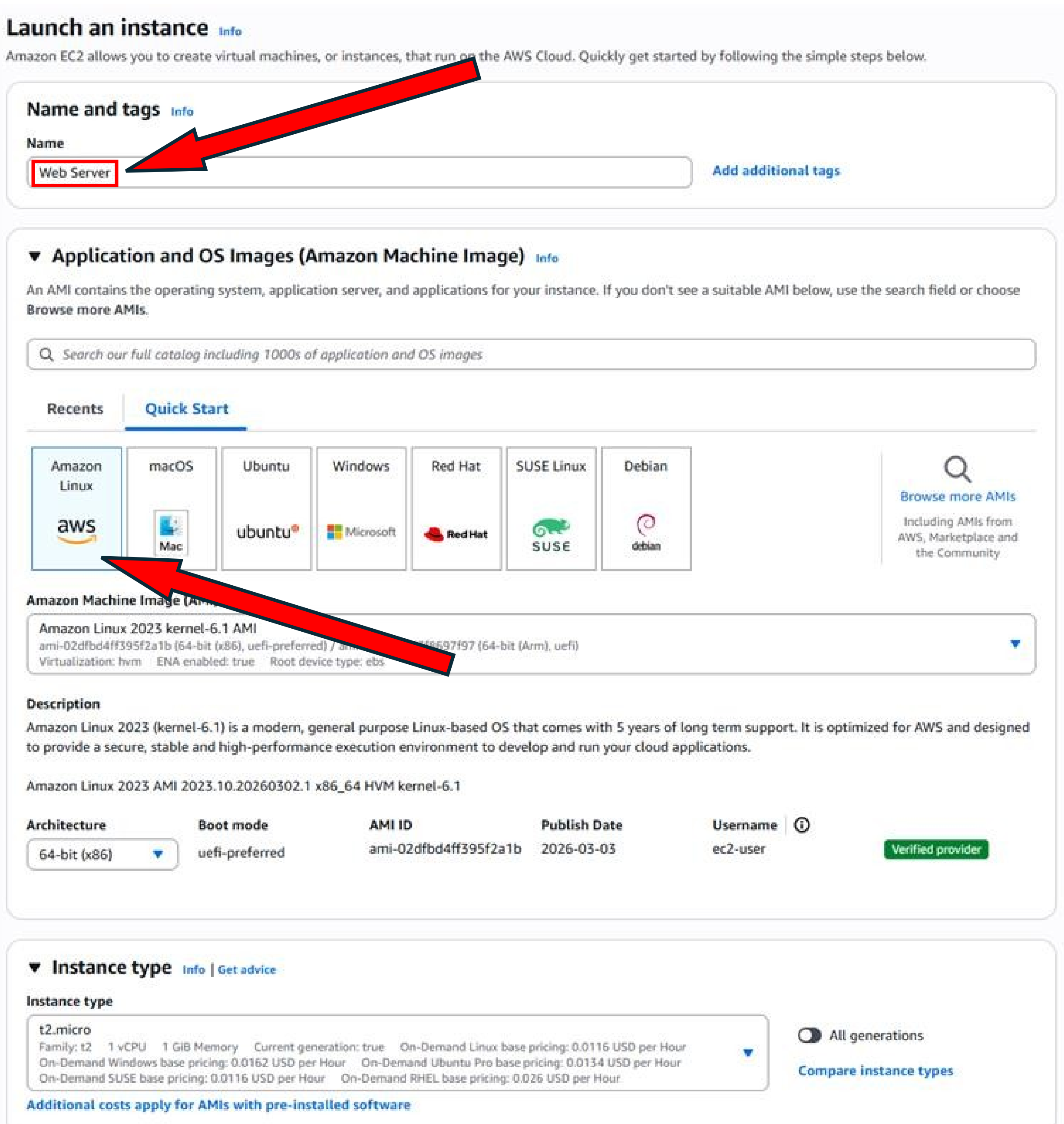
Lab 3:
Go to services and choose “Compute” and choose “EC2”



After that, choose “Launch instance”

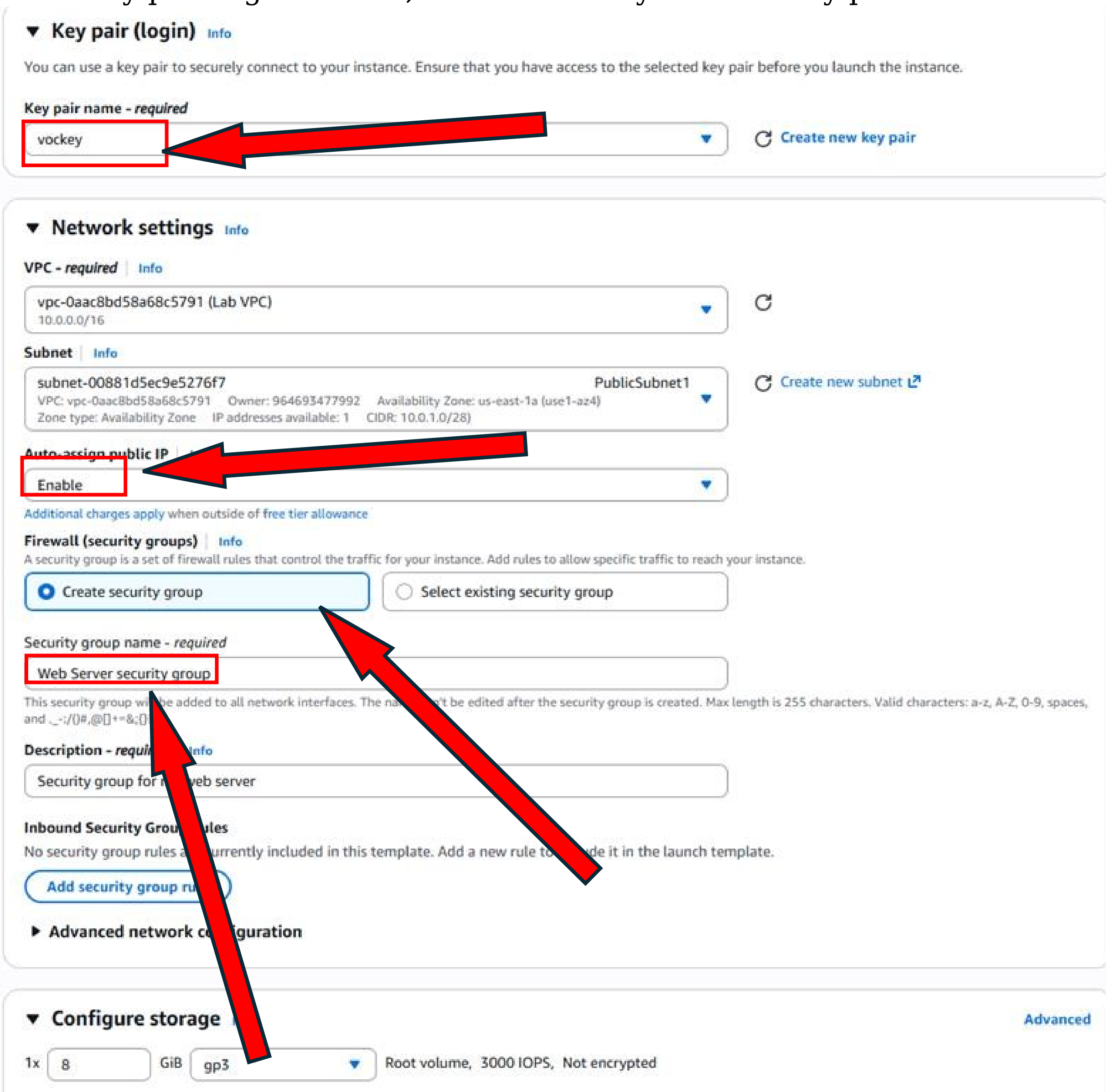


Set the instance name as "Web Server"

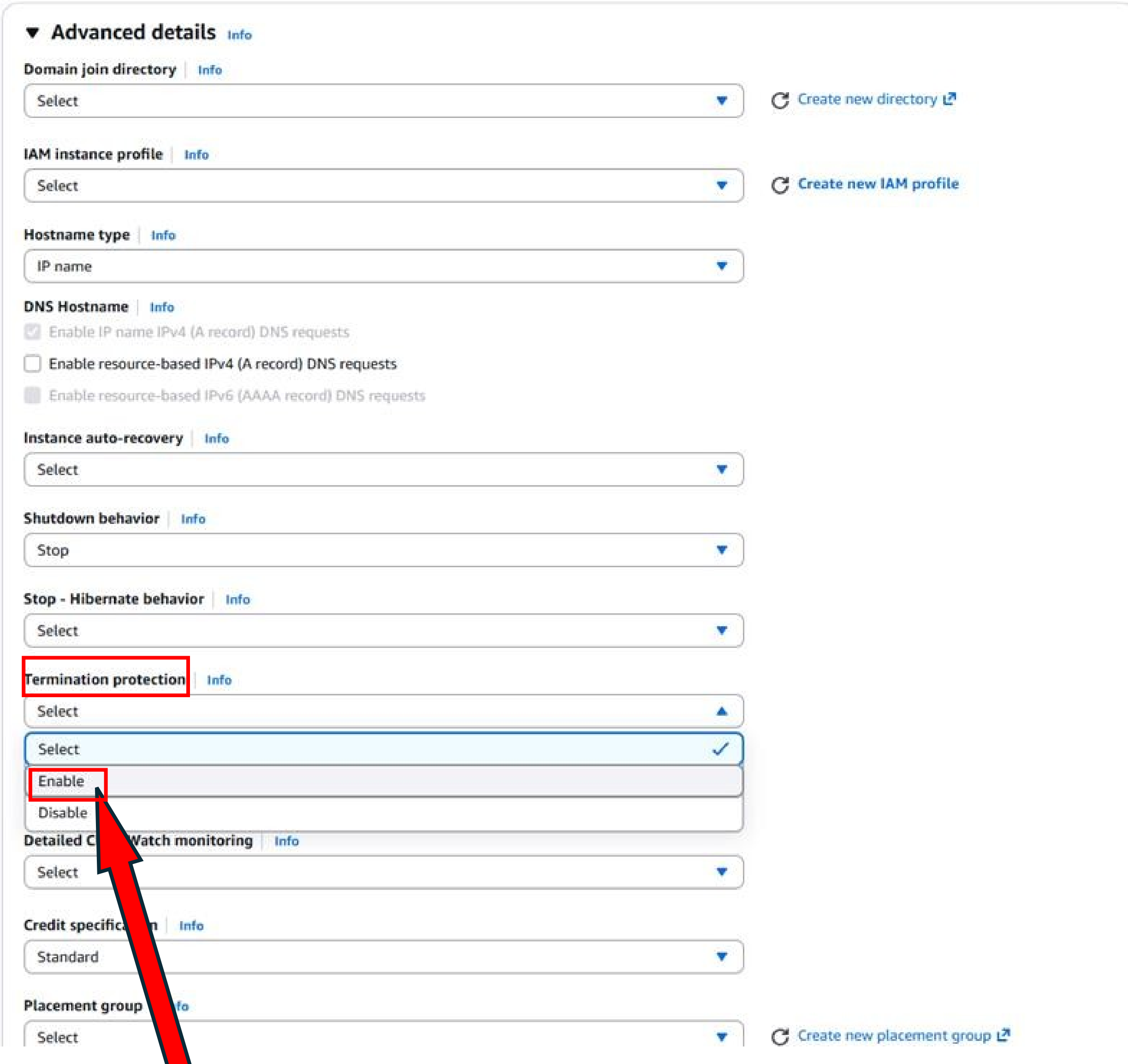


In the Quick Start AMIs, keep default Amazon Linux AMI selected.

In the Key pair login section, choose “vockey” for the Key pair name



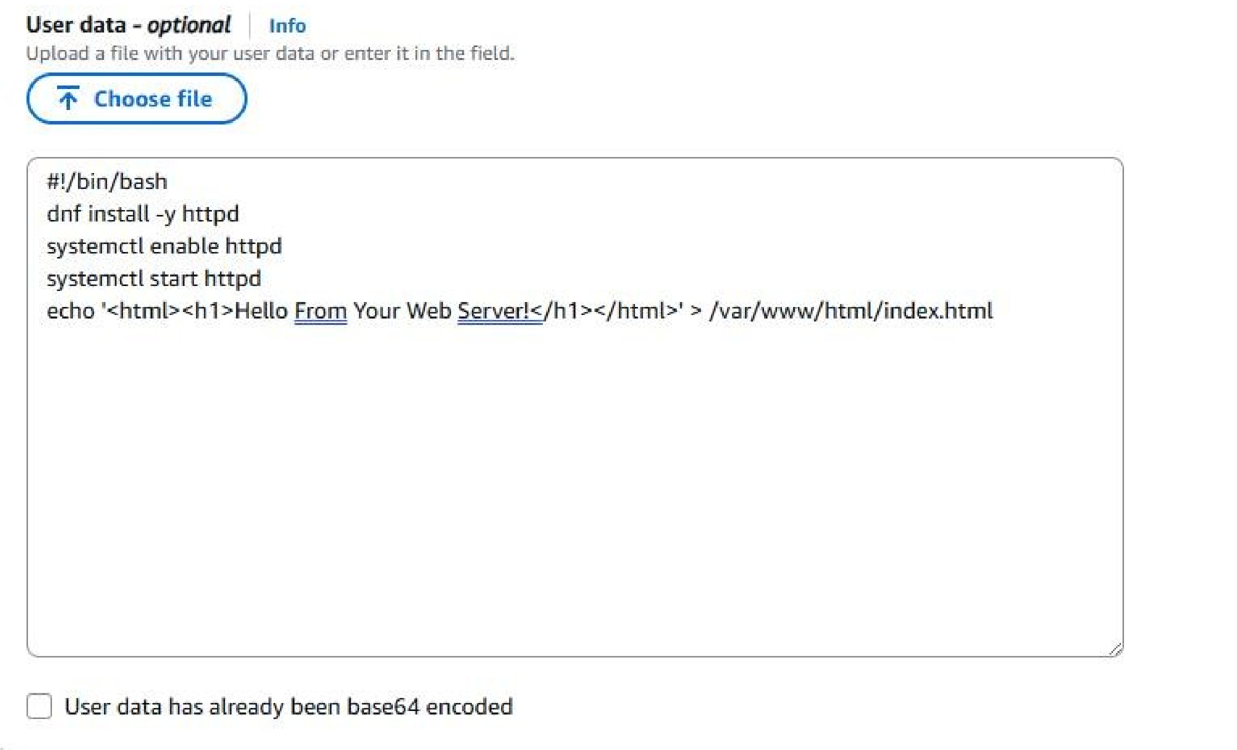
Expand and then go to “Advanced Details”. Select enable for Termination protection



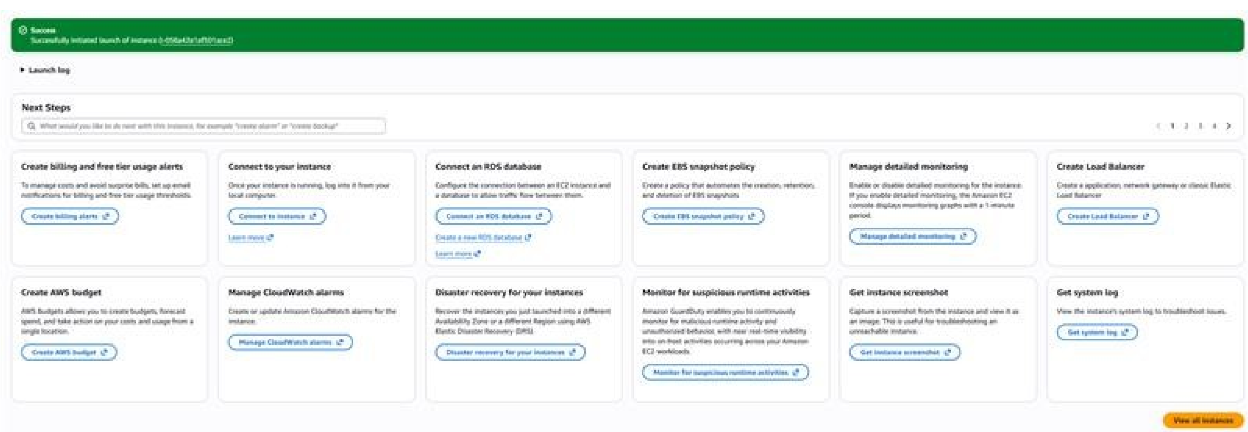
Go all the way to the bottom of the page and copy this code into the "User data" box:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```

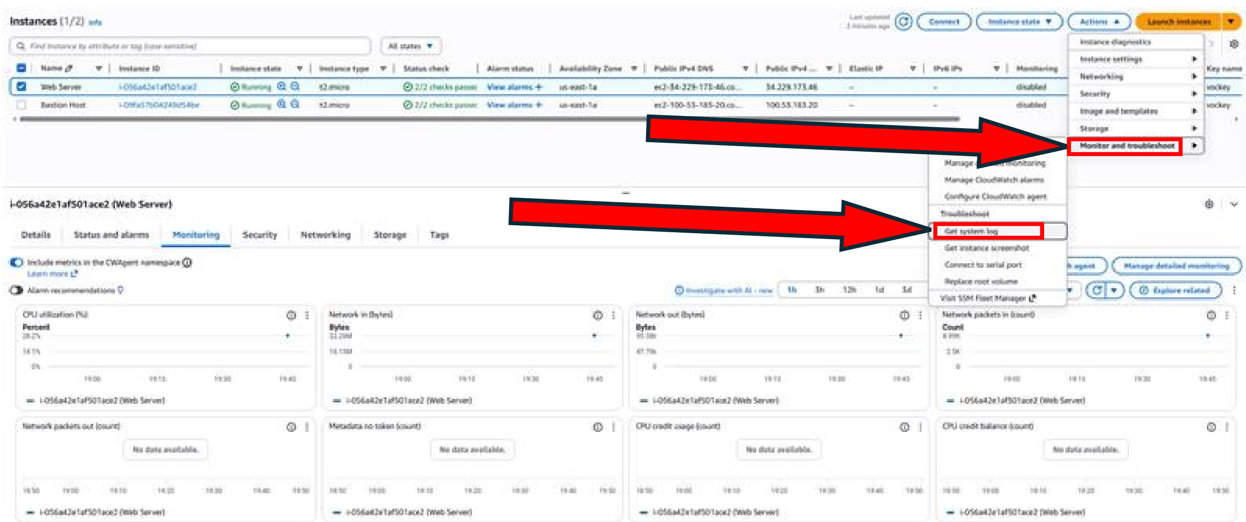
It will look something like:



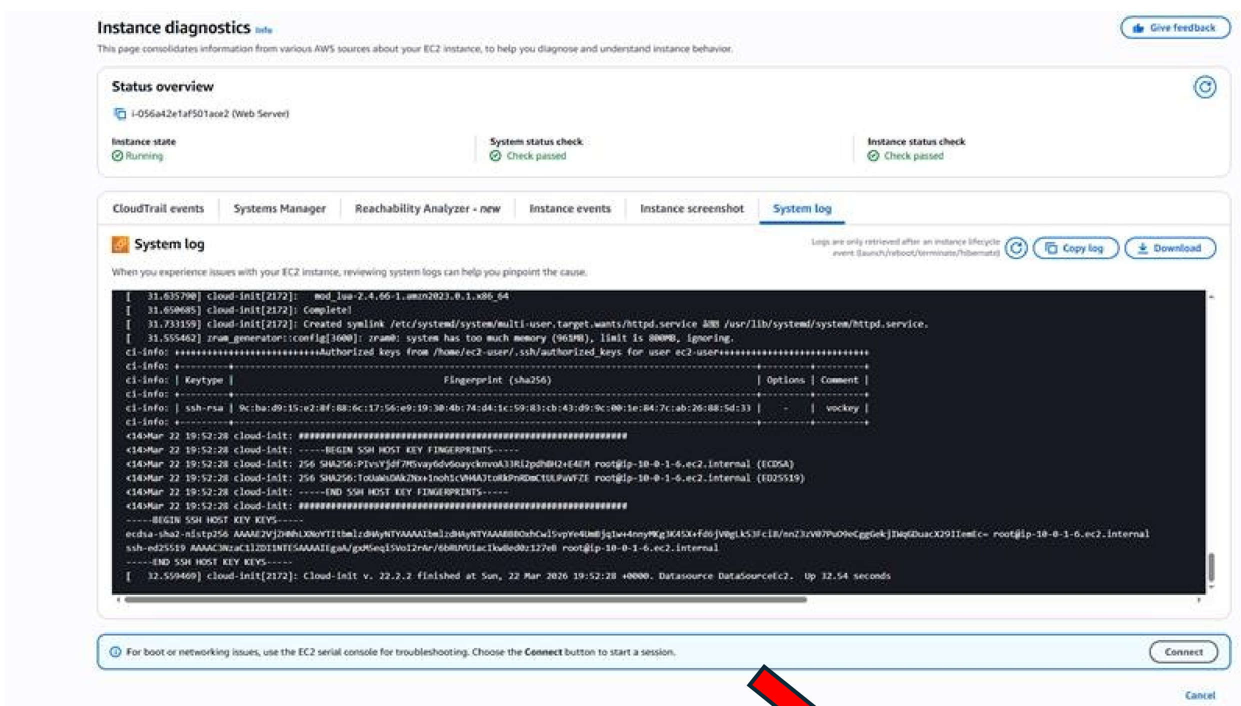
After that, at the bottom of the “Summary” panel choose “Launch Instance”, if everything is done correctly, you will see a green success message at top of the page.



Next, you will monitor your instance, go to the “Monitoring” tab, select enlarge. Then in the “Actions” menu on the top of the console, select “Monitor and troubleshoot” then go to “Get system log.”

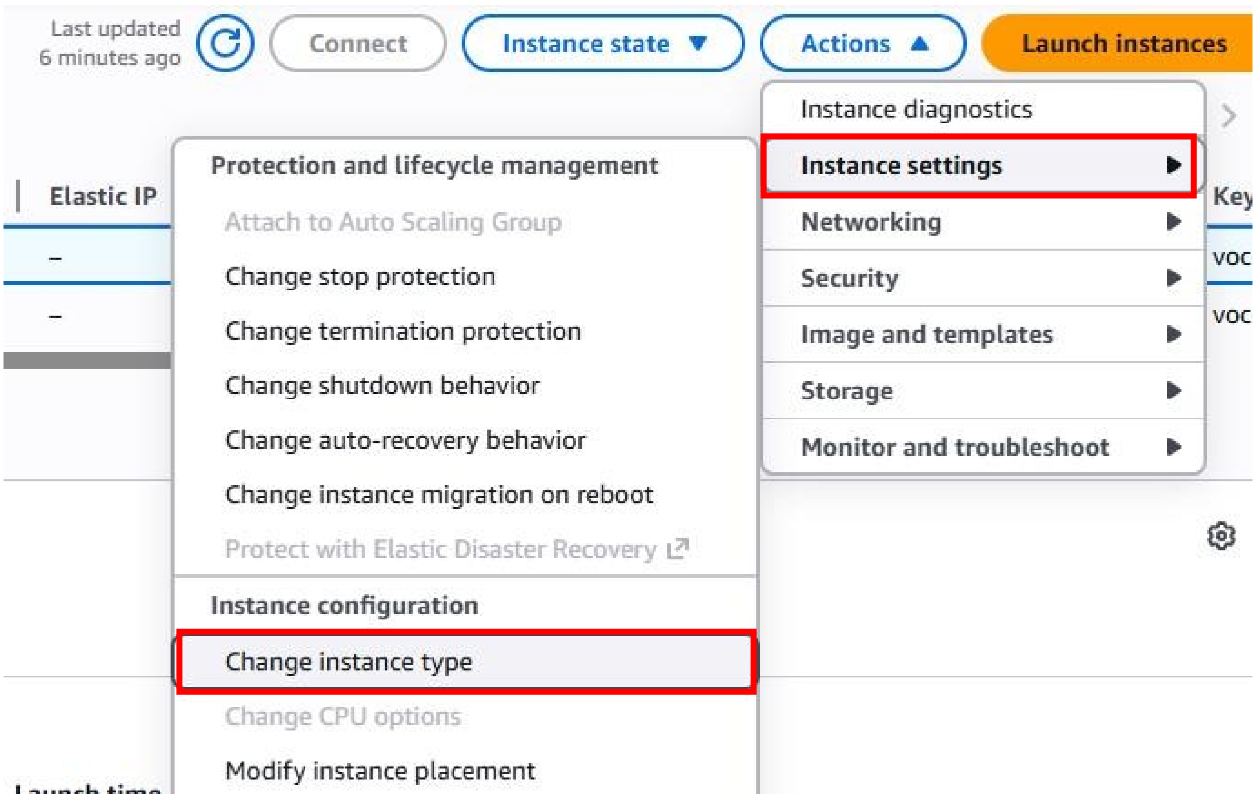


You will then see the user data that you added when creating the instance, through the system log, as shown below:

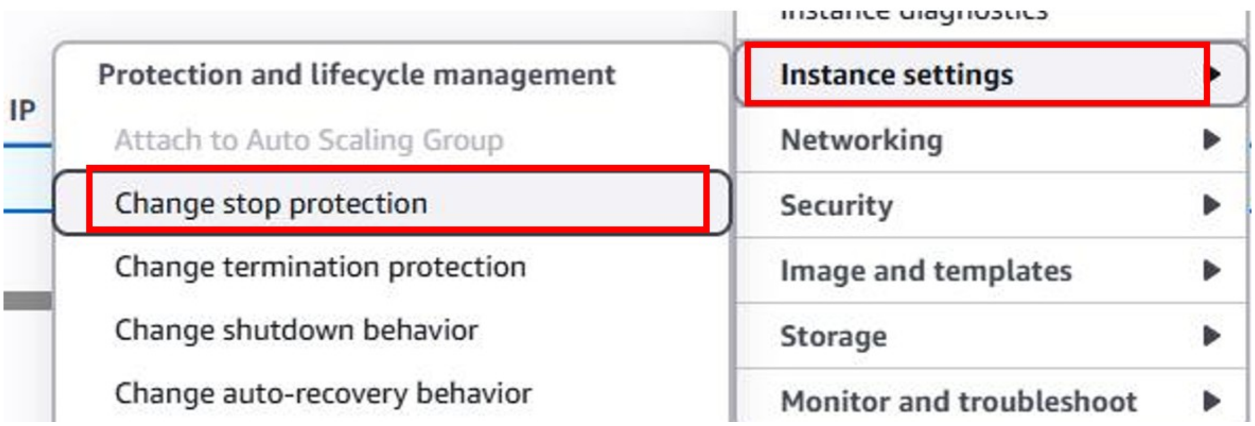


Now you will resize your instance, first stop your instance.

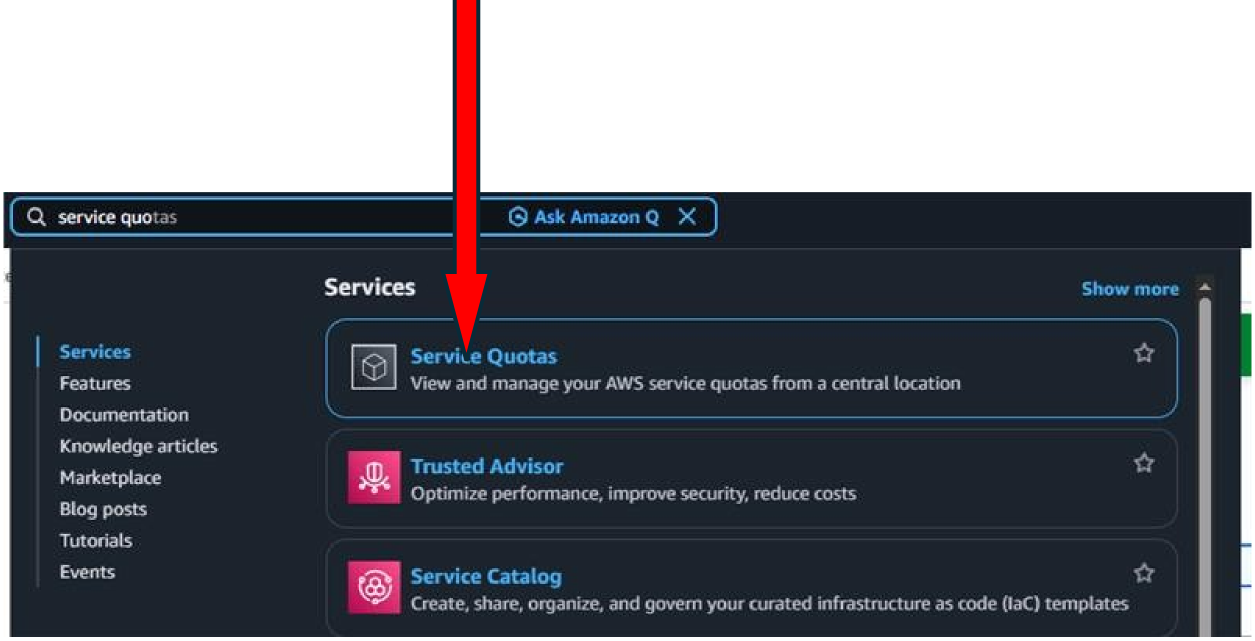
Now you will change the instance type and enable stop protection. Select the “Web Server” interface, and then in the “Auctions” Menu, select “Instance setting”, then “Change instance type”, then configure the instance type to t2.small. Finally click “Apply.



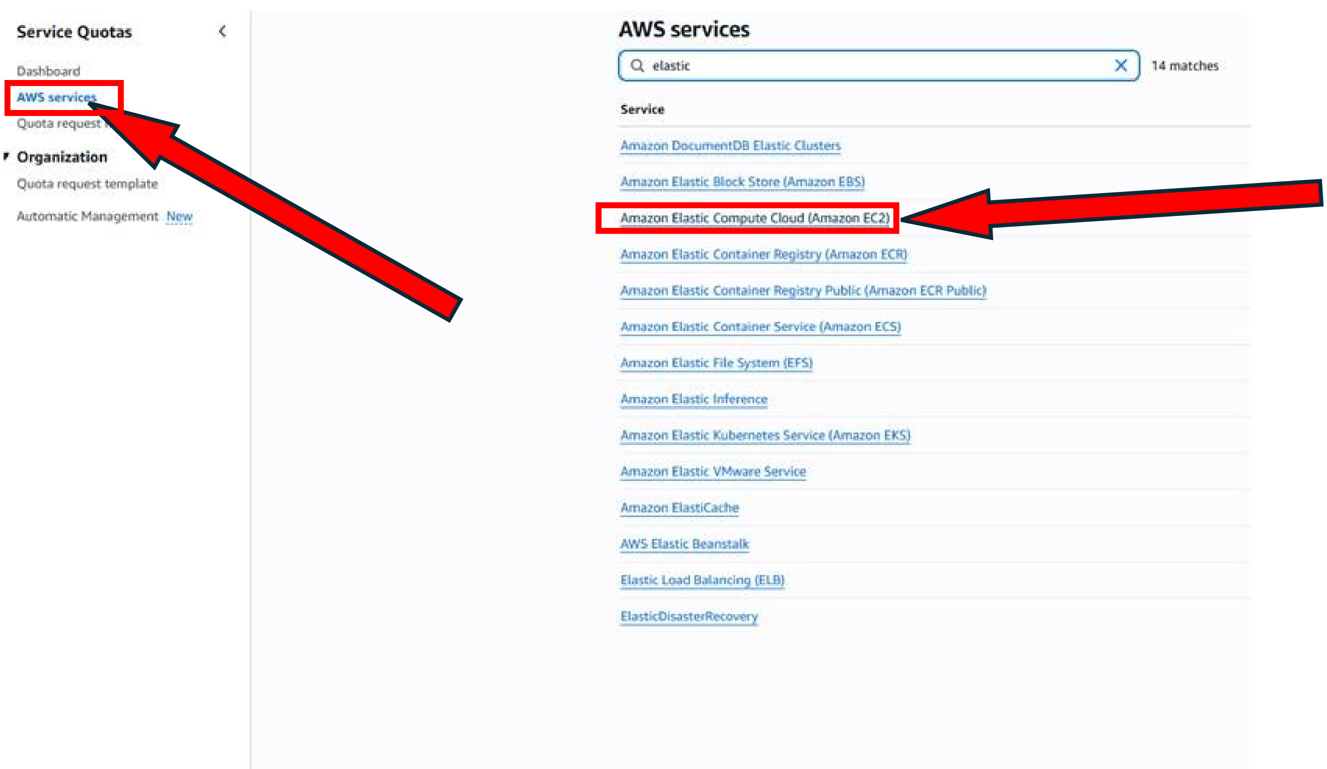
Select the “Web Service” interface again, and go to Actions, and select “Instances settings”, then “Change stop protection”



Next you will explore EC2 limits, in the box next to Services, search and select “Service Quotas”, choose “AWS services”



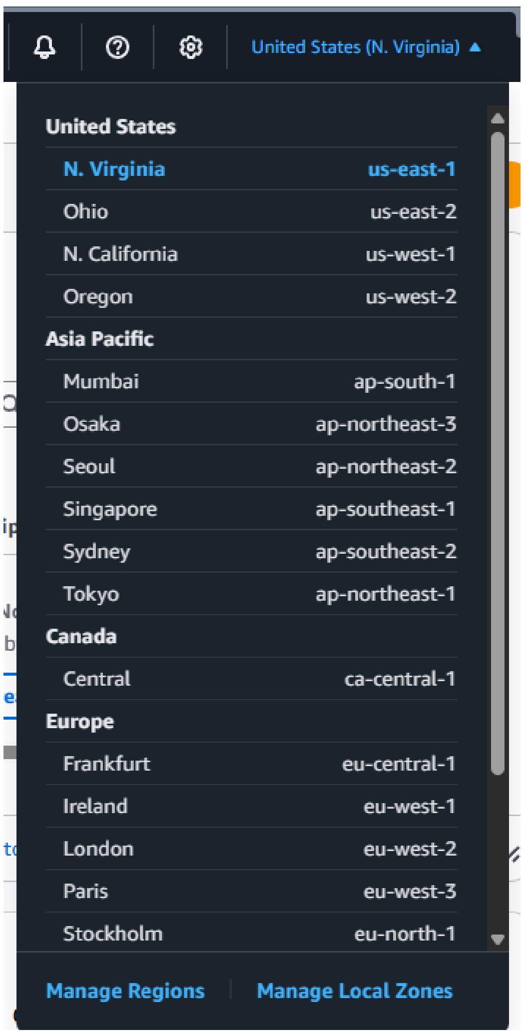
After Selecting Service Quotas, search for “Amazon Elastic Compute Cloud (Amazon EC2)” in AWS services



To test stop protection go to EC2, choose “Instances”, go to “Instance state” and then select “Stop instance”.

Problems:

For the First lab in the section where the user is meant to stop the instance, the instance might not be there when the user logs into the IAM user account.
To fix this issue you need to switch to the region you used at the start of the lab, as shown below:



Conclusion:

Throughout these 3 labs the user learns how to manage users, security credentials, and permissions on AWS resources all through AWS

Identity and Access Management. The labs also demonstrate how Amazon Virtual Private Clouds function and enable the user to launch their own VPC and observe it. Finaly, Amazon Elastic Compute Cloud is also showcased in the final lab where the user sets up failure resilience and can test it themselves.